Table of Content

Module 1: Introduction

1 Project Scenario and Overview

1.1 Introduction to the Commercial Space Age

The commercial space age is here, and several companies are making space travel more affordable and accessible.

Some major companies in the commercial space industry include:

- **Virgin Galactic** – Provides suborbital spaceflights.
- **Rocket Lab** – A small satellite launch provider.
- **Blue Origin** – Manufactures suborbital and orbital reusable rockets.
- **SpaceX** – One of the most successful commercial space companies.

1.2 SpaceX Accomplishments

Some of SpaceX's major accomplishments include:

- Sending spacecraft to the International Space Station (ISS).
- Developing **Starlink**, a satellite internet constellation that provides internet access through satellites.
- Sending manned missions to space.

1.3 Why SpaceX Launches Are Less Expensive

One reason SpaceX has achieved significant success is that its rocket launches are relatively inexpensive.

- A **Falcon 9** launch is advertised at approximately **$62 million**.
- Other launch providers may charge **more than $165 million** per launch.

A major reason for these lower costs is that SpaceX can **reuse the first stage of its rockets**.

Therefore, if we can determine whether the first stage will successfully land, we can estimate the overall cost of a launch.

1.4 Understanding the Falcon 9 Rocket

To understand the scale of the Falcon 9 rocket, diagrams created by **Forest Katsch** at **zlsadesign.com** are used.

Forest Katsch is:

- A 3D artist
- A software engineer
- A creator of infographics related to spaceflight and spacecraft art

- A software developer

1.4.1 Components of Falcon 9

1.4.1.1 Payload

The payload is enclosed inside the **fairings** (protective shells that surround and protect the payload during launch).

1.4.1.2 Second Stage

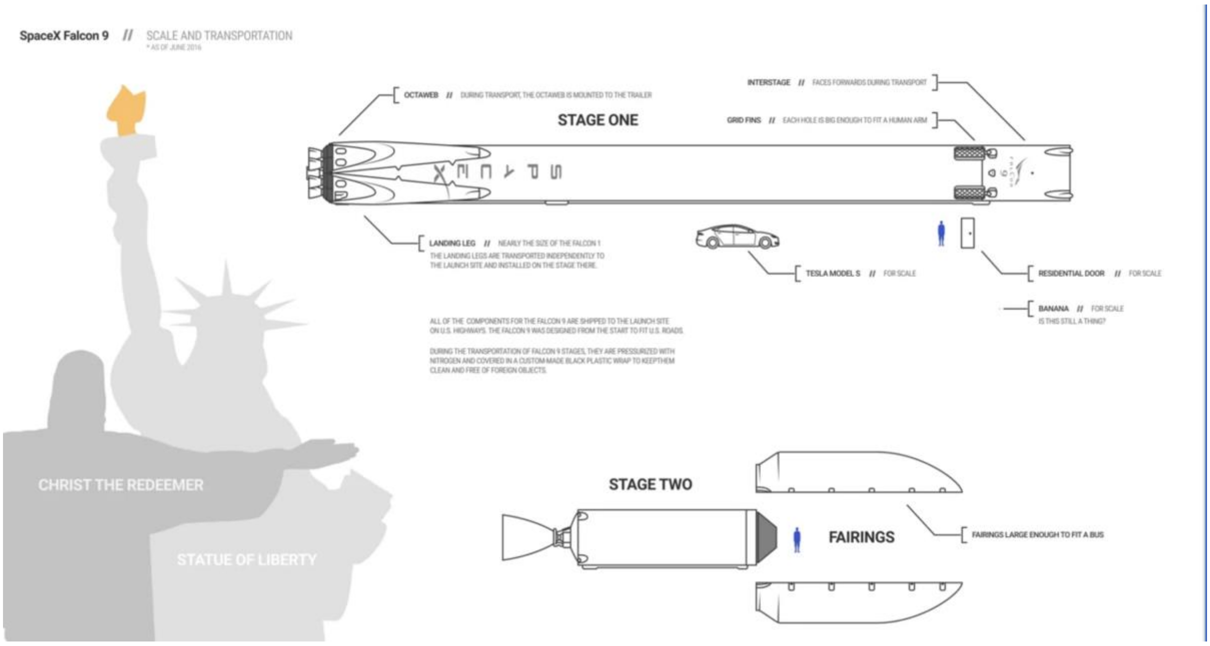
The second stage helps bring the payload into orbit.

1.4.1.3 First Stage

The first stage performs most of the work during launch and is significantly larger than the second stage.

Characteristics of the first stage:

- Very large in size.
- Expensive to manufacture.
- Responsible for most of the rocket's thrust during launch.



1.5 First Stage Recovery

Unlike many other rocket providers, SpaceX can recover the first stage of the Falcon 9 rocket.

However, the first stage does not always land successfully.

Possible outcomes include:

- Successful landing and recovery.
- Crashing during landing.
- Intentional sacrifice of the first stage because of mission requirements.

Mission parameters that may require sacrificing the first stage include:

- Payload requirements
- Target orbit
- Customer requirements

1.6 Capstone Project Scenario

In this capstone project, you will take the role of a **Data Scientist** working for a new rocket company called **Space Y**.

Space Y aims to compete with SpaceX and is founded by billionaire industrialist **Allon Musk**.

1.7 Project Objectives

Your responsibilities in this project include:

1.7.1 Determine the Price of Each Launch

You will gather information about SpaceX and estimate the cost of each rocket launch.

1.7.2 Create Dashboards

You will create dashboards to help your team understand and analyze launch data.

1.7.3 Predict First Stage Reusability

You will determine whether SpaceX will reuse the first stage of its rockets.

Instead of using rocket science principles to determine whether the first stage will land successfully, you will:

- Gather publicly available data.
- Train a machine learning model.
- Use the model to predict whether the Falcon 9 first stage will be reused.

The ability to predict first-stage recovery is important because it directly influences the cost of a rocket launch.

2 Collecting the Data with an API

2.1 Introduction

In this capstone assignment, we work with SpaceX launch data gathered from an API, specifically the **SpaceX REST API**. This API provides information about:

- The rocket used
- Payload delivered
- Launch specifications
- Landing specifications
- Landing outcomes

The primary goal is to use this data to **predict whether SpaceX will attempt to land a rocket or not**.

2.2 SpaceX REST API

The SpaceX REST API base URL is:

```
api.spacexdata.com/v4/
```

Some example endpoints are:

```
/capsules
/cores
```

For this project, we use the following endpoint to obtain historical launch data:

```
api.spacexdata.com/v4/launches/past
```

2.3 Retrieving Data from the API

To access the launch data, we perform a **GET request** using the `requests` library.

```
import requests

response = requests.get('api.spacexdata.com/v4/launches/past')
```

The response can be viewed using:

```
response.json()
```

The API response is returned in **JSON (JavaScript Object Notation)** format, specifically as a **list of JSON objects**, where each JSON object represents a single launch.

2.4 Converting JSON to a DataFrame

To work with the data efficiently in Python, the JSON data needs to be converted into a Pandas DataFrame.

This can be achieved using the json_normalize() function, which converts structured JSON data into a flat table.

```
from pandas import json_normalize

df = json_normalize(response.json())
```

The json_normalize() function "normalizes" the nested JSON structure and transforms it into a tabular format suitable for analysis.

2.5 Alternative Data Source: Web Scraping

Another popular source for obtaining Falcon 9 launch data is through **web scraping** related Wiki pages.

In this project, the **BeautifulSoup** package is used to:

- Scrape HTML tables containing Falcon 9 launch records.
- Parse the data from these tables.
- Convert the extracted information into a Pandas DataFrame for further analysis and visualization.

2.6 Data Wrangling Tasks

The raw data needs to be transformed into a clean and meaningful dataset. The major tasks involved are:

- Wrangling data using an API
- Sampling data
- Dealing with null values

2.7 Gathering Additional Information Using API Endpoints

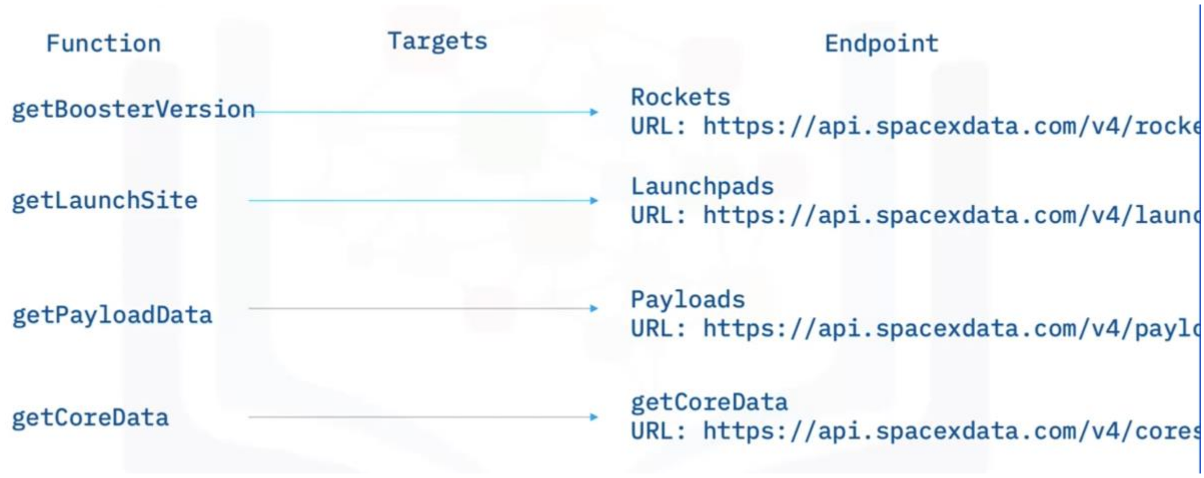
Some columns, such as rocket, contain only an **identification number (ID)** rather than actual information.

This means the API must be queried again using other endpoints to retrieve detailed information for each ID.

The provided functions use the following information:

- Booster
- Launchpad
- Payload
- Core

The retrieved data is stored in lists and then used to create the final dataset.



2.8 Sampling Data

The launch dataset includes information for both:

- Falcon 1
- Falcon 9

However, the project only focuses on **Falcon 9 launches**.

Therefore, the data must be filtered to remove all **Falcon 1** launches.

2.9 Handling Null Values

Real-world data is rarely perfect and often contains **NULL values (missing values)**.

These missing values must be handled to make the dataset suitable for analysis.

2.9.1 Handling NULL Values in PayloadMass

The PayloadMass column contains NULL values.

The required steps are:

1. Calculate the mean of the PayloadMass column.
2. Replace the NULL values with the calculated mean.

Example:

```
mean_payload = df['PayloadMass'].mean()
df['PayloadMass'].fillna(mean_payload, inplace=True)
```

2.9.2 Handling NULL Values in LandingPad

The LandingPad column is intentionally left with NULL values because a NULL value indicates that **no landing pad was used**.

These NULL values will be handled later using:

- One-Hot Encoding (a technique that converts categorical values into numerical columns).

2.10 Key Takeaways

- SpaceX launch data is collected using the SpaceX REST API.
- The endpoint api.spacexdata.com/v4/launches/past provides historical launch information.
- API responses are returned as JSON data.
- json_normalize() is used to convert JSON into a tabular DataFrame.
- BeautifulSoup can be used to scrape additional Falcon 9 launch records from web pages.
- Data wrangling involves:
  - Gathering additional information through API endpoints
  - Filtering out Falcon 1 launches
  - Handling missing values
- NULL values in PayloadMass are replaced with the mean.
- NULL values in LandingPad are preserved and handled later using one-hot encoding.

3 Creating an Effective Data Findings Report

3.1 Introduction

While finding and cleaning data is an important first step in data analysis, a concept can be lost if you are unable to organize and present the findings effectively to your audience.

After the data has been collected, cleaned, and organized, the work of interpretation begins. At this stage, you are able to obtain a complete view of the data and, hopefully, answer the questions that were formed before starting the analysis.

You then begin to compose a **data findings report**, which explains what was learned during the analysis. Depending on the stakeholders (people who have an interest in the project or its results) and how they receive the information, your report may take different forms, such as:

- A paper-style report
- A slideshow presentation
- A combination of both

The findings report is a crucial part of data analysis because it communicates the discoveries and insights obtained from the data.

3.2 Planning the Report

When beginning the report-writing process, the collected data and information may seem overwhelming. The best way to overcome this is by first creating an outline.

Creating an outline helps you:

- Obtain a complete picture of the report.
- Organize your ideas logically.
- Write in a precise yet simple manner.

Although there are many formats for creating a data-driven presentation, a simple outline is often the easiest and most effective approach.

"Always structure the outline for your audience and create a presentation appropriate to the specific situation."

3.3 Structure of a Data Findings Report

A typical data findings report consists of the following sections:

- Cover Page
- Executive Summary
- Table of Contents
- Introduction
- Methodology
- Results
- Discussion
- Conclusion
- Appendix

Note: The depth and length of each section may vary depending on the audience and the format of the report.

3.4 Cover Page

The cover page includes:

- Title of the presentation or report
- Author's name
- Date

3.5 Executive Summary

The executive summary briefly explains the details of the project and should be considered a **stand-alone document** (a section that can be read independently without reading the full report).

Key points:

- Summarizes the main points of the report.
- Information is taken from the report itself.
- Repeating important information is acceptable.
- No new information should be introduced.

3.6 Table of Contents

The table of contents lists all sections and subsections of the report.

Its purpose is to:

- Provide readers with an overview of the report.
- Help readers quickly navigate to the sections that are most important to them.

3.7 Introduction

The introduction explains:

- The nature of the analysis.
- The problem being addressed.
- The questions that the analysis aims to answer.

3.8 Methodology

The methodology section explains:

- The data sources used during the analysis.
- The overall plan for collecting and analyzing the data.
- The analytical techniques or models that were applied.

For example, it may specify whether:

- Cluster analysis was used.
- Regression analysis was used.

3.9 Results

The results section provides detailed information about:

- How the data was collected.
- How the data was organized.
- How the data was analyzed.

This section also includes:

- Charts
- Graphs
- Other visualizations

These visual elements support the findings and highlight the most important or complex results.

By interpreting the data, you provide the audience with a detailed explanation of the findings and demonstrate how they relate to the problem stated in the introduction.

3.10 Discussion

The discussion section explains the findings and their implications (the possible meaning, impact, or consequences of the findings).

This is where you engage your audience by discussing what can be concluded from the research.

Example:

Suppose you conducted research on the top programming languages for college graduates. The discussion might address questions such as:

- Do graduates need to learn multiple programming languages to remain competitive in the job market?
- Will one programming language continue to dominate, or will different languages be important for different careers?

3.11 Conclusion

The conclusion is the final section of the main report.

It should:

- Reiterate the problem presented in the introduction.
- Provide an overall summary of the findings.
- State the outcome of the analysis.
- Mention any future work or additional steps that may be taken.

3.12 Appendix

The appendix contains information that did not fit within the main body of the report but is still important enough to include.

Examples include:

- Locations where the raw data was collected.
- Additional supporting information.
- Resources.
- Acknowledgements.
- References.

3.13 Summary

A well-structured data findings report helps communicate the results of data analysis clearly and effectively. By organizing the report into sections such as the executive summary, introduction, methodology, results, discussion, conclusion, and appendix, the findings become easier to understand, navigate, and present to different audiences.

4 The Report Structure

4.1 Introduction

Before starting the analysis, think about the structure of the report. The structure depends largely on the length and purpose of the document.

- **Brief reports** (typically five or fewer pages) are concise and focus on summarizing the key findings.
- **Detailed reports** (often exceeding 100 pages) gradually build the argument and include detailed discussions of relevant literature, research methodology, data sources, intermediate findings, and final results.

Reports prepared by leading consulting firms such as Deloitte and McKinsey also vary in length depending on their objectives. Brief reports are commonly written as commentaries on current trends or developments that attract public or media attention. In contrast, comprehensive reports provide an in-depth review of the subject, supported by extensive data analysis, commentary, newly collected data, or interviews with industry experts.

Even when preparing a short report, it is recommended to follow a complete report structure that includes all essential sections.

4.2 Recommended Report Structure

A well-organized report should include the following components:

- Cover Page
- Table of Contents
- Executive Summary (or Abstract)
- Introduction
- Literature Review
- Methodology
- Results
- Discussion
- Conclusion
- References
- Acknowledgments
- Appendices (if required)

4.3 Cover Page

The cover page is one of the most frequently overlooked parts of a report. This omission is not limited to undergraduate students; even doctoral candidates and professionals sometimes forget to include it. Many reports published by well-known organizations also lack a proper cover page.

At a minimum, the cover page should contain:

- Title of the report

- Name(s) of the author(s)
- Author affiliation(s)
- Contact information
- Name of the institutional publisher (if applicable)
- Date of publication

The publication date is particularly important because reports without a publication date cannot be properly cited. Including contact information also makes it easier for readers to communicate with the author.

4.4 Table of Contents

A **Table of Contents (ToC)** is like a map for a journey that has never been taken before. It provides readers with an overview of the document before they begin reading.

The ToC helps readers by:

- Showing the main sections of the report.
- Listing tables and figures.
- Providing a preview of the document's organization.

A Table of Contents should always be included when the main content of the report (excluding the cover page, ToC, and references) is five pages or longer.

4.5 Executive Summary (Abstract)

Even for short reports, it is recommended to include an **Abstract** or an **Executive Summary**.

A good executive summary:

- Explains the main arguments of the report.
- Summarizes the key findings.
- Is typically limited to three paragraphs or fewer for short reports.
- May be longer for comprehensive reports consisting of hundreds of pages.

4.6 Introduction

The introductory section prepares readers for the report by introducing the topic and defining the problem being addressed.

This section is especially helpful for readers who are unfamiliar with the subject, allowing them to understand the context before moving into more detailed discussions.

4.7 Literature Review

The literature review examines previous research related to the topic.

Its length depends on the complexity of the research area.

- If previous research largely agrees on the topic, only the most influential studies need to be cited.
- If the research contains differing opinions, caveats, or ongoing debates, a more detailed literature review is necessary.

The literature review also serves to:

- Identify existing knowledge gaps.
- Explain why further research is needed.
- Introduce the research questions.
- Present the research hypothesis.

4.8 Methodology

The methodology section explains how the research was conducted.

This section should describe:

- Research methods used.
- Data sources.
- Data collection procedures (if new data was collected).
- Reasons for selecting particular variables, datasets, and analytical methods.

The methodology should also explain how these choices help answer the research questions, often by referring back to findings from the literature review.

4.9 Results

The results section presents the empirical findings of the analysis.

Typically, the presentation begins with:

- Descriptive statistics (summary statistics describing the data).
- Illustrative graphics such as charts, plots, or maps.

The analysis may then proceed to formal hypothesis testing.

Depending on the nature of the study, additional analytical techniques may include:

- Regression models.
- Categorical data analysis.
- Time-series analysis.
- Data mining techniques.

In many business reports, statistical details are minimized, while charts and visualizations are emphasized to make the findings easier for non-technical audiences to understand.

4.10 Discussion

The discussion section interprets the findings presented in the results section.

This is where narrative becomes important, helping the numerical results communicate the overall argument of the report.

The discussion typically:

- Revisits the original research questions.
- Refers back to the knowledge gaps identified in the literature review.
- Explains how the findings contribute to filling those gaps.
- Interprets the significance of the results.

Not every analysis produces a complete or definitive answer. In many cases, the findings only partially answer the research questions and must be interpreted alongside important limitations and caveats (limitations or conditions that affect the interpretation of the results).

4.11 Conclusion

The conclusion generalizes the specific findings of the study and highlights their broader significance.

Rather than focusing on limitations, this section emphasizes the value and contributions of the research.

A strong conclusion should:

- Summarize the major findings.
- Reinforce the importance of the research.
- Discuss potential practical applications.
- Suggest directions for future research and development.

4.12 References

The references section lists all sources cited throughout the report.

Proper referencing:

- Gives credit to original authors.
- Supports the credibility of the report.
- Enables readers to locate the cited sources.

4.13 Acknowledgments

The acknowledgment section recognizes individuals or organizations that contributed to or supported the work.

This may include:

- Academic supervisors.
- Colleagues.
- Funding organizations.
- Institutions.
- Other contributors.

Acknowledging support is considered good academic and professional practice.

4.14 Appendices

Appendices contain supplementary material that supports the report but would interrupt the flow of the main discussion if included within the primary sections.

Examples include:

- Large tables.
- Questionnaires.
- Survey forms.
- Additional figures.
- Detailed calculations.
- Supporting documents.

Appendices should only be included when necessary.

5 Delivering Effective Data-Driven Presentations

5.1 Introduction

After spending weeks or even months studying the data, the time comes to report your findings. The questions have been answered, and the story is ready to be shared. The challenge is presenting those findings in a way that ensures the audience understands and remembers the intended message.

Delivering data-driven presentations may seem straightforward, but there are several important factors to consider to accurately convey your message and keep the audience engaged.

5.2 Best Practices for Delivering Data-Driven Presentations

When creating and delivering a presentation, keep the following principles in mind:

- Make sure charts and graphs are not too small and are clearly labeled.
- Use the data only as supporting evidence.
- Share only one key point from each chart or graph.
- Eliminate data that does not support the key message.

5.3 Ensure Charts and Graphs Are Easy to Read

Small charts and labels can easily be overlooked, making the presentation difficult to follow.

To ensure readability:

- Test the visualizations by sitting at different distances, similar to where your audience will be seated.
- If the data cannot be seen clearly, consider redesigning the visualization by increasing its size, improving labels, or simplifying the chart.

5.4 Use Data to Support the Story, Not Tell the Entire Story

When preparing a report, it may seem logical to fill the slides with data because the findings are based on extensive analysis. However, while this approach may make sense from a data analyst's perspective, the audience is more likely to see a collection of numbers rather than understand the message.

A better approach is to:

- First identify the key messages that need to be communicated.
- Build the presentation as a story around those messages.
- After creating the outline, insert the data to support each finding.

By using data as supporting evidence instead of the main focus, the presentation becomes more engaging, easier to follow, and more interesting for the audience.

5.5 Present One Key Message per Visualization

Charts and graphs are excellent tools for communicating information, but providing too much information in a single visualization can confuse the audience.

For example, a pie chart containing numerous categories and multiple messages makes it difficult for viewers to determine:

- The key message.
- The point the presenter is trying to convey.
- Where the audience should focus their attention.

Instead, each chart or graph should communicate only one primary idea. Avoid combining multiple messages into a single visualization, as doing so may create confusion and reduce the effectiveness of the presentation.

5.6 Eliminate Unnecessary Data

Data analysts often spend months researching and exploring data. During this process, some findings may seem interesting but are not relevant to the project's objectives.

Including unnecessary details can weaken the overall presentation and distract the audience from the main message.

To keep the presentation effective:

- Remove data that does not support the key message.
- Highlight only the data points that reinforce your main ideas.
- Keep the presentation clear, concise, and focused.

5.7 Key Takeaways

An effective data-driven presentation should:

- Use clear and readable charts and graphs.
- Treat data as supporting evidence rather than the entire presentation.
- Communicate only one key message through each visualization.
- Remove irrelevant information that does not support the main message.
- Keep the presentation clear, concise, engaging, and easy for the audience to understand.

5.8 Conclusion

A successful data-driven presentation is not about showing every piece of data collected during the analysis. Instead, it is about telling a compelling story supported by relevant data. By focusing on clear visuals, emphasizing one idea at a time, and eliminating unnecessary information, you can effectively engage your audience and deliver a clear, memorable message.